

ELSENBORN-BUTGENBACH AP, BelgiumWMO: **064960**Lat: **50.4822N**Lon: **6.1814E**Elev: **570**StdP: **94.66**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.4	-14.3	1.2	-8.5	-11.3	1.5	-6.7	11.0	5.8	9.9	4.3	1.8	70	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	27.3	18.0	25.1	17.1	23.2	16.4	19.0	24.9	18.0	23.4	17.1	21.8	2.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.0	21.1	16.1	12.3	19.8	15.2	11.6	18.9	56.3	25.0	53.1	23.3	50.2	22.0	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.6	-16.0	30.6	3.5	1.7	-18.5	31.8	-20.5	32.8	-22.5	33.7	-25.0	34.9		
(5)			-16.3	21.6	3.4	1.2	-18.8	22.4	-20.8	23.1	-22.7	23.7	-25.2	24.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	0.1	0.6	3.3	6.6	10.4	13.2	15.3	14.7	11.5	8.3	4.1	1.2		
	DBStd	6.48	4.16	4.26	3.76	3.82	3.61	3.44	3.23	3.13	2.99	3.60	3.70	3.98		
	HDD10.0	1487	306	264	209	116	39	7	1	2	17	76	178	274		
	HDD18.3	3984	564	498	467	353	246	157	106	120	206	310	426	532		
	CDD10.0	570	0	0	1	13	52	104	164	148	61	24	2	0		
	CDD18.3	24	0	0	0	0	1	4	11	8	1	0	0	0		
	CDH23.3	400	0	0	0	2	17	69	156	142	14	1	0	0		
	CDH26.7	86	0	0	0	0	2	11	37	36	1	0	0	0		
(14) Wind	WSAvg	3.0	3.6	3.5	3.3	3.0	2.9	2.7	2.5	2.3	2.6	3.0	3.2	3.4		
(15) Precipitation	PrecAvg	1049	106	89	79	65	75	82	89	92	81	80	94	117		
	PrecMax	1349	256	258	167	169	168	163	211	206	228	229	156	302		
	PrecMin	767	5	14	11	0	27	24	34	28	11	27	5	16		
	PrecStd	150	54	52	47	36	31	36	39	48	50	41	37	59		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	12.4	17.0	22.7	25.6	28.4	30.4	30.2	25.5	20.7	14.7	10.4		
		MCWB	8.6	6.7	9.1	12.8	16.0	18.1	18.6	19.2	16.9	14.3	11.7	9.0		
	2%	DB	8.2	9.2	14.3	19.6	22.1	25.3	27.3	26.9	22.5	17.7	12.0	9.2		
		MCWB	7.3	6.4	8.2	11.7	14.3	17.2	18.2	18.1	16.0	13.6	10.2	8.2		
	5%	DB	6.9	7.6	11.5	16.6	20.2	22.9	24.9	24.4	19.9	15.4	10.5	8.0		
		MCWB	6.1	6.2	7.4	10.4	13.3	16.3	17.2	17.2	14.8	12.6	9.3	7.3		
	10%	DB	5.4	6.3	9.3	14.1	17.9	20.6	22.8	21.9	17.4	13.7	9.3	6.6		
		MCWB	4.7	5.2	6.8	9.2	12.5	15.1	16.6	16.0	13.8	11.8	8.3	5.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	8.6	10.6	13.8	17.5	20.0	20.2	20.4	17.9	15.7	12.0	9.7		
		MCDB	9.5	10.0	14.0	20.2	22.6	25.8	26.6	27.0	22.9	19.3	13.2	10.0		
	2%	WB	7.5	7.2	9.3	12.4	15.7	18.2	19.0	18.8	16.7	14.2	10.5	8.4		
		MCDB	8.1	8.4	12.3	18.0	20.8	23.1	25.1	24.9	21.1	16.5	11.6	8.9		
	5%	WB	6.1	6.3	8.2	10.9	14.1	17.0	18.1	17.7	15.5	13.1	9.5	7.4		
		MCDB	6.7	7.3	10.6	15.1	18.6	21.6	23.5	22.9	19.1	15.0	10.4	8.0		
	10%	WB	4.8	5.2	7.1	9.8	12.9	15.7	17.1	16.7	14.3	12.1	8.4	6.0		
		MCDB	5.3	6.2	9.1	13.5	16.9	19.7	21.9	20.8	16.9	13.6	9.2	6.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.9	5.8	7.8	9.9	10.1	10.2	10.2	10.6	9.3	7.3	4.9	4.6		
		MCDBR	5.5	8.5	13.3	15.3	14.8	15.2	15.4	15.8	14.2	10.8	7.2	5.4		
	5% WB	MCWBR	5.0	6.0	8.1	8.4	8.0	8.1	7.4	7.9	8.4	6.8	5.5	4.7		
		MCWBR	4.9	6.3	10.2	13.2	12.7	13.2	13.2	13.7	12.5	9.0	6.1	4.6		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.316	0.347	0.379	0.387	0.389	0.389	0.383	0.356	0.336	0.312	0.291		
		taud	2.455	2.402	2.352	2.294	2.306	2.313	2.304	2.347	2.410	2.467	2.468	2.438		
	Ebn at Noon		753	822	854	861	870	868	861	848	836	788	720	696		
		Edh at Noon	59	80	100	119	123	123	122	112	94	75	58	52		
(44) All-Sky Solar Radiation	RadAvg		0.77	1.45	2.59	4.12	4.85	5.33	5.05	4.33	3.24	1.89	0.89	0.59		
	RadStd		0.12	0.28	0.39	0.48	0.47	0.50	0.60	0.44	0.39	0.18	0.12	0.09		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page